Inverse design of broadband highly reflective metasurfaces using neural networks

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Metamaterials exhibit optical properties not observed in traditional materials. Such behavior emerges from the interaction of light with precisely engineered subwavelength features built from different constituent materials. Recent research into the design and fabrication of metamaterial-based devices has established a foundation for the next generation of functional materials. Of particular interest is the all-dielectric metasurface, a two-dimensional metamaterial exploiting shape-dependent resonant features while avoiding losses through the use of dielectric building blocks. However, even this simple metamaterial class has a nearly infinite number of possible configurations; researchers now require new methods to efficiently explore these types of design spaces. In this work, we employ rigorous coupled wave analysis to calculate reflection and transmission spectra associated with a class of open-cylinder all-dielectric metasurface. By altering the geometric parameters of open-cylinder metasurfaces, we generate a sparse training data set and construct artificial neural networks capable of relating metasurface geometries to reflection and transmission spectra. Here, we successfully demonstrate that pseudo autodecoder neural networks can suggest device geometries based on a requested optical performance—inverting the design process for this metasurface class. As an example, we query for and discover a particular open-cylinder metasurface displaying a reflection band $R \geqslant 99\%$ centered at $\lambda_0 = 1550\,\text{nm}$ that is much broader $\Delta\lambda = 450\,\text{nm}$ than anything reported for optical metasurfaces. We then analyze the modal interplay in the open-cylinder metasurface to better understand the underlying physics driving the broadband behavior. Ultimately, we conclude that neural networks are ideally suited for generally approaching these types of complex inverse design problems.

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I. INTRODUCTION

Metamaterials exhibit novel properties due to fine-tuned light-matter interactions with subwavelength scatterers. Even though such devices were theorized long ago [1], only recently have devices with feature sizes relevant to infrared and visible wavelengths become experimentally viable [2]. To date, various bottom-up and top-down fabrication approaches have driven the design and fabrication of these structures depending on the constituent materials in use and the geometrical arrangements sought after [3–8]. Metamaterials demonstrate potential as compact optical components in a variety of applications, including the design and manufacture of plasmonic metasurfaces [9,10], solar sails [11], optical absorbers [12,13], near-perfect reflectors [14–16], and beam collimators [17]. Simpler metamaterials have intriguing optical properties as well. The two-dimensional metasurface, for example, is a subclass composed of a single patterned layer and has been investigated for a variety of applications [18–20]. As a particular example, the all-dielectric metasurface [21–25] can possess desirable performance with regards to highly reflective low-loss devices and can even be harnessed in tunable systems [26,27].

Even a relatively simple metasurface has a near-infinite design space in terms of the possible ways to configure the constituent materials and surface geometry. This is further complicated by the incoming radiation defined by its elevation and azimuthal angle of incidence, polarization, beam shape, wavelength, and even nonlinear interactions [28]—all of which affect the response of the metasurface device. Even with the most modern high-performance computing resources and optimized computational electromagnetic methods (CEMs), one simply cannot exhaustively search the massive parameter spaces in question. Because a comprehensive "brute-force" search is untenable, any attempts at creating a metamaterial with a targeted performance must leverage an understanding of the underlying physics. For example, Mie scattering concepts [15,29,30], Fano resonance behaviors [6,24,26,31], and adjoint methods [32] have been used to design particular subsets of device geometries. However, there exists no prescription for inverting Maxwell's equations; thus, one cannot generally design an arbitrary metamaterial starting with targeted optical constraints. In a few limited cases, traditional optimization routines are capable of designing metasurfaces with desired optical properties [33–36]; however, each new targeted property and/or device geometry must be handled individually, ultimately limiting the overall efficiency of the discovery process. As an example, genetic algorithms can be implemented to fine-tune design parameters but tend to struggle in higher-dimensional searches, often getting stuck in the local minima of the large parameter space. As a result, efficiently approaching this inverse design problem continues to be a subject of considerable interest [37].

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Machine learning (ML) methods have recently risen in prominence to more judiciously handle the metasurface inverse design problem [7,38–43]. Artificial neural networks (ANNs) [44] are particularly well suited for this task because they are *function approximators*: a neural network learns a multifunctional relationship $\vec{y} = f(\vec{x})$ when trained on input data $\vec{x}$ and output data $\vec{y}$. If an inverse relationship between the input and output data exists, the neural network may instead be trained to learn the inverse relationship $\vec{x} = f^{-1}(f(\vec{x})) = g(\vec{y})$ [45]. In fact, recent work combining CEMs and ML has now provided a pathway for exploring these kinds of higher-dimensional inverse problems [7,38,39,46].

In this work, we consider a system of nanoscale Si open cylinders [47] on a SiO_2 substrate. This geometry is a generalized version of a recently investigated all-dielectric solid-cylinder metasurface with an experimentally demonstrated broadband near-perfect reflection in the short-wave infrared regime ($R > 98\%$ across a $\Delta\lambda = 200$ nm wide bandwidth covering $\lambda_0 = 1550$ nm) [16]. With this previous work serving as a precedent, this paper seeks to demonstrate the power of ANNs for inverse metamaterial design by querying for an all-dielectric open-cylinder metasurface that out performs the previously considered solid-cylinder metasurface in terms of both broadband reflection and peak reflectivity. We calculate the reflection and transmission spectra of open-cylinder metasurfaces randomly sampled from a four-dimensional (4D) input parameter space using rigorous coupled-wave analysis (RCWA). The output of these simulations serves as training data sets for ANNs built to suggest an inverse design solution. The most predictive success is observed with an ANN trained in a pseudo autodecoder (PAD) neural network architecture [48]. We then utilize the PAD-trained neural network to uncover a highly reflective, broadband open-cylinder metasurface centered around the telecommunications wavelength $\lambda_0 = 1550$ nm exhibiting $R \geqslant 99\%$ over twice the bandwidth $\Delta\lambda = 450$ nm previously reported. Following the work of Ref. [22], we then analyze the modes arising within the open-cylinder metasurface to better understand the origin of the highly reflective broadband behavior. We conclude that although this method could be used, it becomes untenable with increasing metasurface complexity. Thus, the ML protocols defined in this work provide a more robust and general method in designing metasurfaces, avoiding the need to derive *ad hoc* solutions for individual geometries. The PAD ANN approach accelerates the overall design process and enables the rapid discovery of tailored functional materials.

II. METASURFACE ANNs: TRAINING AND ARCHITECTURE

The metasurface class explored here consists of a periodic array of Si open cylinders on a SiO_2 substrate surrounded by air, as illustrated in Fig. 1. The computational electromagnetic method of choice is RCWA since it is well suited for simulating such a periodic device [22]. RCWA efficiently models the Floquet boundary conditions in the x and y directions while avoiding any approximations to Maxwell's equations. This allows for computationally inexpensive extraction of metasurface modes and reflection and transmission coefficients. We use the open source S^4 [49] RCWA package to simulate our metasurfaces coupled with the SIGNAC data

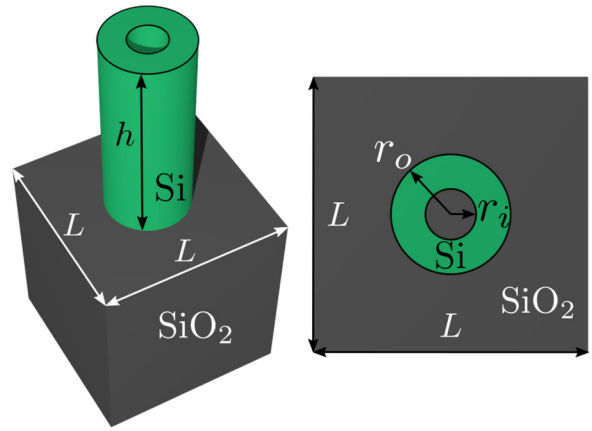


FIG. 1. Top and angled views of a unit cell for an open-cylinder metasurface design defined with Floquet boundary conditions in the transverse directions, x and y, with finite layers in the z direction. This example design has an outer radius $r_o = 211$ nm, inner radius $r_i = 89$ nm, height $h = 942$ nm, and lattice spacing $L = 964$ nm.

management framework [50] to account for the large amount of data required to adequately train ANNs. See the Rigorous Coupled Wave Analysis section in the Supplemental Material for more information regarding the software we developed and employed to create synthetic training data sets [51].

Here, we restrict the RCWA simulations to normally incident, TE-polarized light in the range $\lambda = [800\,\text{nm}, 4000\,\text{nm}]$ with 1-nm resolution, resulting in reflection and transmission spectra containing $N_R = N_T = 3201$ data points per metasurface design. Designs are randomly sampled from the 4D design space shown in Table I, with 10-nm resolution for a total design space of approximately 5×10^8 possible open-cylinder arrangements. To prevent the simulation of unphysical devices (e.g., devices in which the outer edge of open cylinders overlap), the maximum values of $r_{o,\max}$ and $r_{i,\max}$ are further restricted: $r_{o,\max} \leqslant 0.4L$, $r_{i,\max} \leqslant 0.8r_o$, and $r_i < r_o$. This reduces the total design space to sample to 1.3×10^7 possible open-cylinder arrangements. In total, we generate two separate training sets, one with square lattices and the other with hexagonal lattices. The lattice spacing defines the primitive vectors: $L:\ \vec{a} = [L, 0], \vec{b} = [0, L]$ for square lattices and $L:\ \vec{a} = [L, 0], \vec{b} = [L\cos\pi/3, L\sin\pi/3]$ for hexagonal lattices. Each set consists of roughly 9000 randomly sampled open-cylinder designs. These simulations use $N_{\text{spatial harmonics}} = N_G = 15^2$, maximizing throughput while still accurately capturing trends contained in the reflection and transmission spectra. Later on, when devices of interest are chosen, we resimulate those candidates with an increased number of spatial harmonics ($N_G = 21^2$) to more accurately capture the reflection and transmission behavior. (See the Rigorous Coupled Wave Analysis section in the Supplemental Material for more information regarding the choice of N_G [51].)

TABLE I. Ranges of values sampled during RCWA simulation of nanoscale silicon open cylinders.

r_o (nm)	r_i (nm)	h (nm)	L (nm)
120–2000	80–1900	200–1000	300–2200

The reflection and transmission data, paired with the associated metasurface design as described by the 4D vector $(r_o,\ r_i,\ h,\ L)$, form the training and validation data sets for our ANNs. Because we seek to find a telecom band reflector, we down sample the total number of data points in both the reflection and transmission spectra to $N_{\mathrm{R}} = N_{\mathrm{T}} = 512$ values in the range of [1300 nm, 1800 nm] for targeted reflection and transmission at $\lambda_0 = 1550$ nm. All data sets are additionally scaled to the range [0, 1] to maximize the performance of the ANNs with existing training algorithms. The reflection and transmission spectra are already in this range; however, the metasurface design variables $(r_o,\ r_i,\ h,\ L)$ must be scaled by their maximum considered values: $(r_o^* = \frac{r_o}{r_{o,\max}})$, $(r_i^* = \frac{r_i}{r_{i,\max}})$, $(h^* = \frac{h}{h_{\max}})$, $(L^* = \frac{L}{L_{\max}})$. We note that the choice of 9000 designs is somewhat arbitrary but sufficient for our case: the ANNs successfully converge during training and predict metasurface designs within the parameter space with high fidelity (coefficient of determination $R^2 \approx 1$). To the authors' knowledge, no studies currently exist which connect the minimum amount of metasurface training data needed to the size of the input parameter space.

We construct and train the ANNs with our curated data on a single Nvidia RTX-2080Ti GPU using the KERAS [52] library with the TENSORFLOW [53] back end. We reserve 10% of the data for performance evaluation after training, reserving an additional 10% for validation during training, using a batch size of 1024. The ANN is trained using the KERAS implementation of the ADAM optimizer [52,54] using the default values for the ADAM hyperparameters (learning_rate = 0.001, beta_1 = 0.9, beta_2 = 0.999, amsgrad = False) and with the mean square error loss function. The ANNs train for up to 2000 generations with early stopping criteria $\Delta\text{loss} < 10^{-9}$. All ANNs utilized in this work use dense layers with the rectified linear unit (ReLU) activation function, except for the use of the sigmoid activation function in the final layer, limiting the output to the range [0, 1].

We start by considering an ANN that predicts the reflection and transmission spectra associated with a given open-cylinder metasurface, i.e., a spectra predicting network (SPN). The architecture of the SPN is described by Eq. (1):

$$4 - 8N_{\mathrm{R}} - 6N_{\mathrm{R}} - 4N_{\mathrm{R}} - 2N_{\mathrm{R}} - \begin{cases} N_{\mathrm{R}} - N_{\mathrm{R}}, \\ N_{\mathrm{T}} - N_{\mathrm{T}}, \end{cases} \tag{1}$$

where the values in Eq. (1) represent the number of neurons at the layer, the dashes indicate different layers, with the first layer appearing on the leftmost part of the equation, and the bracket indicates a fork in the neural network topology. The SPN is shown schematically in Fig. 2(a), providing further clarification of the notation in Eq. (1). (See the Artificial Neural Network Architecture, Training, and Performance section in the Supplemental Material for the KERAS-generated architecture of the SPN [51].) These SPNs take the input device designs in the form of $(r_o^*,\ r_i^*,\ h^*,\ L^*)$ and pass it through four dense layers before splitting into two separate output branches to independently handle the prediction of the reflection and transmission spectra, shown in Fig. 2(b). In principle, this input layer can be arbitrarily large to accompany more complex parameter spaces, keeping in mind that

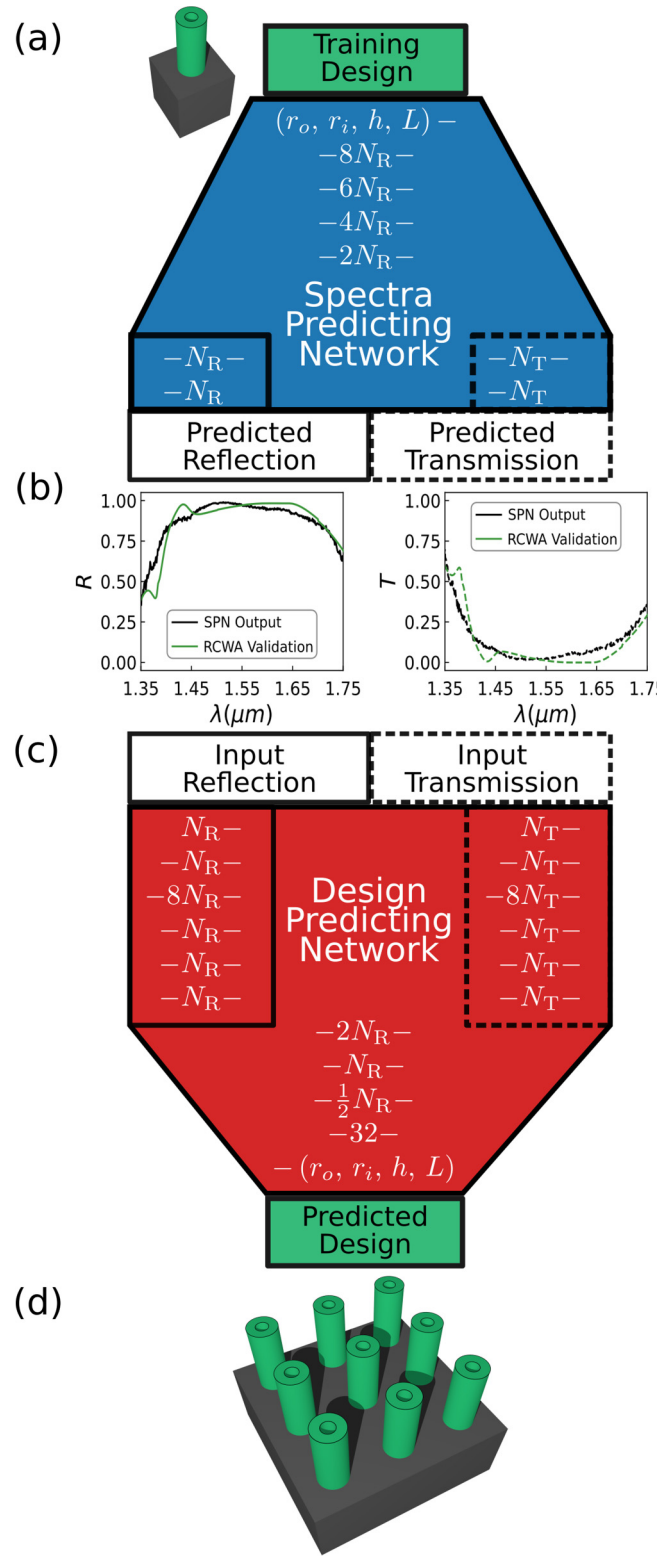


FIG. 2. (a) SPN architecture. The SPN is trained to rapidly predict the expected reflection and transmission spectra. (b) A comparison of the RCWA-generated spectra (green) and the SPN-predicted spectra (black) for reflection (solid) and transmission (dotted) shows general trend agreement. (c) DPN architecture. The DPN may be trained independently with the raw RCWA simulation data (IDPN) or as part of a pseudo autodecoder (PDPN). (d) Open-cylinder metasurface predicted by the DPN based on input spectral data.

larger parameter spaces will require larger training data sets to converge.

There are two essential advantages of training an SPN. The first advantage is with sufficient training data, the SPN begins to coarsely mimic the computational outputs of RCWA for the range of open cylinders considered in Table I. This is true mimicry in the sense that the SPN does not really learn RCWA but instead is able to draw overarching mathematical patterns which correlate the input parameter values with the output reflection and transmission spectra. This important distinction makes the spectral outputs of the SPN better suited for solving the inverse design problem. The second advantage is an adequately trained SPN is able to compute additional output spectra many times faster than the original CEM (in our case, the SPN is about 1 million times faster than RCWA). As long as the training data are sufficient, these two features are ultimately what allow ANNs to converge for much larger parameter spaces.

Second, we create an ANN with the purpose of correlating reflection and transmission spectra with open-cylinder designs: a design-predicting network (DPN). The DPN architecture is described by

$$\left.\begin{matrix} N_{\mathrm{R}} - N_{\mathrm{R}} - 8N_{\mathrm{R}} - N_{\mathrm{R}} - N_{\mathrm{R}} - N_{\mathrm{R}} \\ N_{\mathrm{T}} - N_{\mathrm{T}} - 8N_{\mathrm{T}} - N_{\mathrm{T}} - N_{\mathrm{T}} - N_{\mathrm{T}} \end{matrix}\right\} \\ -2N_{\mathrm{R}} - N_{\mathrm{R}} - \tfrac{1}{2}N_{\mathrm{R}} - 32 - 4. \qquad (2)$$

The DPN architecture is illustrated in Fig. 2(c). (See the Artificial Neural Network Architecture, Training, and Performance section in the Supplemental Material for the KERAS-generated architecture of the DPN [51].) Simulated spectra and the associated metasurfaces form the input and output of the DPN, respectively. This DPN may be trained either independently with the raw RCWA simulation data (IDPN) or with the assistance of the SPN shown in Fig. 2(b), forming a pseudo autodecoder DPN (PDPN). We call the SPN + DPN configuration a pseudo autodecoder due to its resemblance to pseudo autoencoder architecture [38,48]. The SPN layers are frozen during the training of the DPN in the PAD configuration. The two input layer branches of the DPN take in respective reflection and transmission spectra. These spectra pass through five dense layers before merging and compressing through four more dense layers of decreasing size. The final layer contains four output nodes representing the device design in the form of $(r_o^*,\ r_i^*,\ h^*\ L^*)$, shown in Fig. 2(d).

III. ANN RESULTS AND USE FOR INVERSE DESIGN

Performance is averaged across multiple replicas of each ANN. Training ANNs is an inherently stochastic process stemming from the use of stochastic gradient descent in the ADAM optimizer [54], along with the random assignment of data to training and validation data sets. This results in ANNs with slightly different weights and slightly different performances each time an ANN is trained. In every case, the SPN, IDPN, and PDPN network architectures all successfully converge, as conveyed by the validation loss function vs generations plot shown in Fig. 3(a). The predictive powers of both the IDPN and PDPN are then evaluated by harnessing the

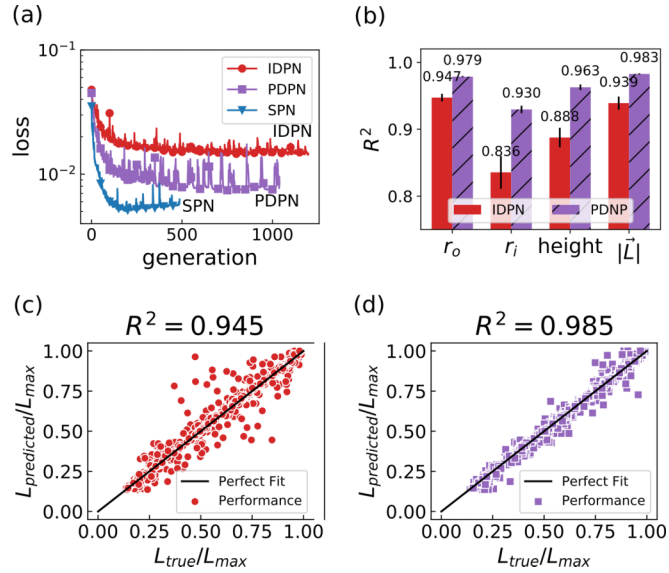


FIG. 3. (a) Loss value comparison for single replicas of the simulator network (blue), the IDPN (red), and the PDPN (purple). All three networks successfully converge, with the PDPN outperforming the IDPN. (b) Performance comparison of the IDPN (red) and PDPN (purple) on validation data sets averaged across ten replicas. The input and output validation data more closely correlate in the PDPN than in the IDPN (error bars are determined by $\mathrm{err} = \sigma_{R^2}/\sqrt{10}$). (c) and (d) Lattice regression performance plot for single DPN replicas of (c) the IDPN and (d) the PDPN, demonstrating superior performance of the PDPN.

aforementioned 10% of data set aside for validation (the SPN validation is implicitly shown in the PDPN validation because it was used to train it). By plotting the actual device parameters in the validation data against the predicted values output by the IDPN and PDPN, the better network architecture is revealed by the coefficient of determination R^2. A comparison for all design parameters is shown in Fig. 3(b), with examples of regression plots for the lattice spacing shown in Fig. 3(c) for the IDPN and Fig. 3(d) for the PDPN. (See Fig. S4 in the Supplemental Material for all regression plots [51].) For all parameters in question, the PDPN outperforms the IDPN in predictive power and also has fewer outliers on its regression.

Previous work [38] suggested that while the functional relationship between the device geometry and the resulting reflection and transmission performance spectra is guaranteed to be one to one, the inverse does not necessarily hold:

$$(R,\ T) = f(r_o,\ r_i,\ h,\ L),$$
$$(r_o,\ r_i,\ h,\ L) \neq f(R,\ T). \qquad (3)$$

This is known as the many-to-one problem. In the case of metasurface design, different metasurfaces may produce similar reflection and transmission spectra, and thus, an ANN may struggle to learn a functional relationship capable of differentiating these geometries. Although we found that the IDPN [Fig. 2(c)] was able to converge, it struggled to consistently predict device geometries from targeted performance spectra (Figs. 3(a) and 3(b); see Fig. S4 for the complete regression performance of the DPN [51]). This is exemplified by the outliers seen in Fig. 3(c), as well as the lack of fidelity

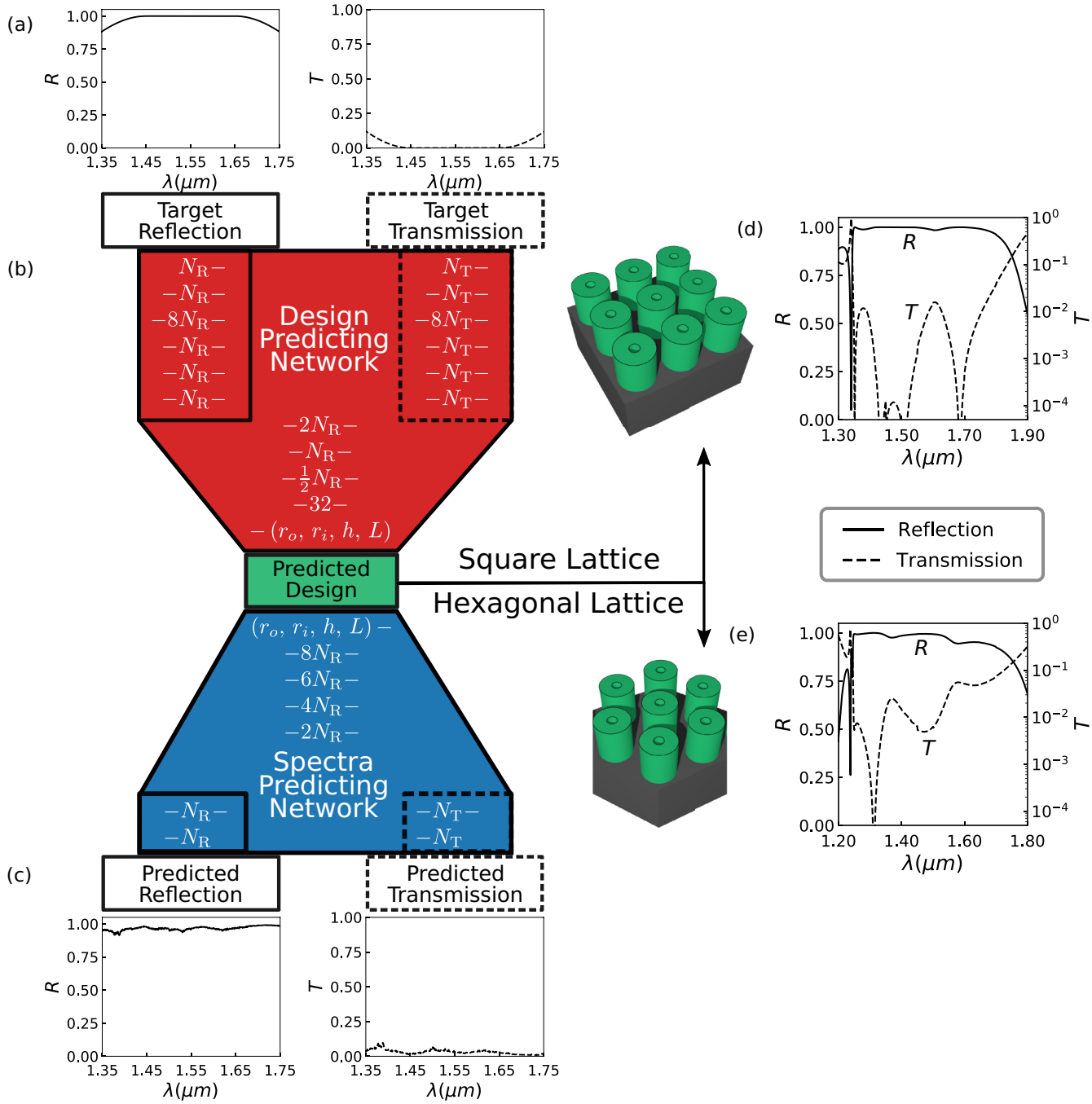


FIG. 4. Inverse design pipeline for open-cylinder metasurfaces with targeted performance. (a) User-input targeted broadband reflection and transmission spectra centered around $\lambda_0 = 1550$ nm. (b) Inverse ANN architecture formed by "flipping and stitching" together the already-trained PDPN and SPN. (c) SPN output indicating the predicted performance of the open-cylinder metasurface designed by the PDPN, showing good agreement with the core features of the target spectra. Together, the PDPN and SPN predict both the metasurface design and resulting reflection and transmission spectrum based on the desired input spectra. High-resolution RCWA verification of (d) the square lattice open-cylinder metasurface and (e) the hexagonal lattice.

in predicting input design variables such as the inner radius, $R^2 = 83.6\%$. Fortunately, we were able to improve upon this via the assistance of the SPN shown in Fig. 2(a), forming a PAD architecture [38,48]. In this latter case, the PAD input will be the same as the PAD output. Without the pretrained SPN, the PAD would learn an arbitrary high-dimensional representation of the device design; instead, we can force the PAD to learn a specific high-dimensional representation of the device design (the reflection and transmission spectra) by coupling the output of the pretrained SPN to the input of the DPN and allowing only the DPN layers of the PAD update during training, i.e., the pretrained SPN layers are "frozen" during training. By forcing the entire PAD to directly associate the input device geometry with the output device geometry, we greatly reduce the number of variables over which to minimize loss ($N_R + N_T \to 4$), improving the ability of the

DPN to accurately predict device designs, giving an average coefficient of determination $R^2 = 96.4\%$ [for all parameters in question, the PDPN outperforms the IDPN in predictive power while also having fewer outliers in its regression; Figs. 3(b)–3(d)]. This improvement is strictly attributable to the fact that the output spectral data from the SPN are built upon fitting functions. These mathematical fits capture only the general trends connecting the input parameter space to the output spectra, allowing the DPN to negotiate the overall patterns instead of each individual spectral value. To describe this from another perspective, consider the finer spectral details observed in the RCWA runs [e.g., Fig. 2(b)] which often have oscillations around an average value. If the DPN is trained on many sets of such data, it will try to account for these features, ultimately overcorrecting the general trends in an attempt to capture these fluctuations.

The PAD not only delivers a SPN and PDPN combination with incredible predictive capability but also provides a way to interpret the predictions of an ANN via the SPN portion of the network like the pseudo autoencoder considered by Liu *et al.* [38]. By inverting and stitching together the now trained PDPN and SPN networks, we can create an inverse design pipeline: a user-defined spectrum [Fig. 4(a)] is first supplied to the PDPN [Fig. 4(b)], designing a device with reflection and transmission spectra that best meet the target criterion. The PDPN-predicted design is then fed into the SPN, delivering a near-instantaneous prediction of the device's performance, as shown in Fig. 4(c). It is important to realize that the PDPN will not necessarily prescribe a design that *exactly* fits the user-defined spectrum. Instead, it will deliver a design that both *exists* within the parameter space on which it was trained and *best fits* the target features: one is guaranteed to obtain an open-cylinder design within the range of Table I whose reflection and transmission features are closest to targeted performance. This guarantee proves to be extremely useful when the training data ranges are preconstrained by fabrication tolerances.

The utility of this inverted PAD network is demonstrated by the choice of target spectra shown in Fig. 4. By querying for an open-cylinder square lattice device with broadband high reflectivity centered around $\lambda_0 = 1550\,\text{nm}$, a candidate is found with device parameters $r_o = 279\,\text{nm}$, $r_i = 77\,\text{nm}$, $h = 593\,\text{nm}$, $|\vec{L}| = 758\,\text{nm}$. When studying this predicted device with a high-resolution RCWA simulation, we find that the predicted performance meets and even exceeds the target criteria: $\Delta\lambda \geqslant 450\,\text{nm}$, $\lambda_0 = 1550\,\text{nm}$, $\Delta\lambda/\lambda_0 > \pm 12.5\%$ with individual reflection peaks with $T \leqslant 10^{-4}$ [Fig. 4(d)]. This same procedure is repeated with the hexagonal lattice training data. A similar broadband device is uncovered, achieving $T(1.25\,\mu\text{m} \leqslant \lambda \leqslant 1.75\,\mu\text{m}) < 10^{-1}$ with geometry $r_o = 259\,\text{nm}$, $r_i = 72\,\text{nm}$, $h = 571\,\text{nm}$, $|\vec{L}| = 743\,\text{nm}$.

IV. MODAL ANALYSIS

In this section, we analyze the eigenmodes within the cylinder geometry studied in Ref. [16] ($r = 221\,\text{nm}$, $h = 502\,\text{nm}$, $L = 679\,\text{nm}$) and a particular high-performance open-cylinder metasurface geometry designed by the PDPN ($r_o = 247\,\text{nm}$, $r_i = 106\,\text{nm}$, $h = 804\,\text{nm}$, $L = 870\,\text{nm}$) to better understand the origin of the highly reflective broadband behavior of both simulated cases. Such modal information can be extracted from the full electromagnetic fields computed with RCWA [55]:

$$\Psi(z) = We^{-\vec{\lambda}Z}\vec{c}, \tag{4}$$

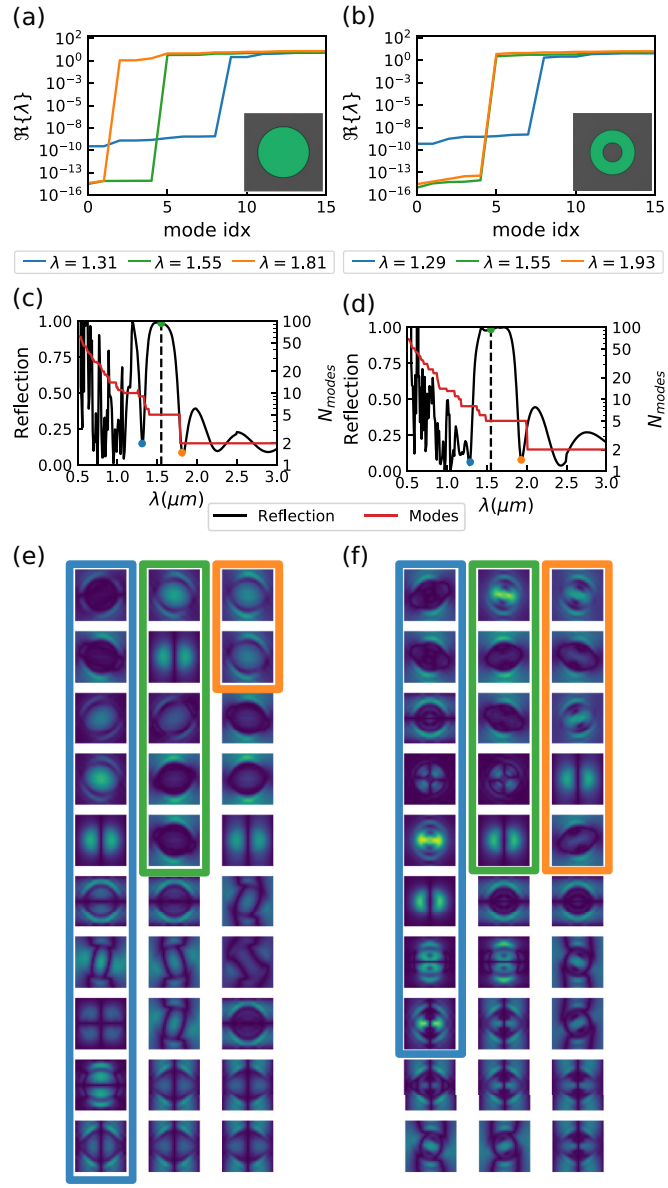


FIG. 5. Comparison of (a), (c), and (e) solid cylinders with geometry $r = 221\,\text{nm}$, $h = 502\,\text{nm}$, $L = 679\,\text{nm}$ and (b), (d), and (f) open cylinders with geometry $r_o = 247\,\text{nm}$, $r_i = 106\,\text{nm}$, $h = 804\,\text{nm}$, $L = 870\,\text{nm}$. (a) and (b) The real component of the first 15 eigenmodes $\text{Re}\{\lambda_i\}$ for three different wavelengths in the reflection spectra for solid and open cylinders. $\text{Re}\{\lambda_i\}$ is plotted on a log scale to show the significant difference in values between the first few eigenmodes and the rest of the eigenmodes. Insets show the top view of the unit cell. (c) and (d) Comparison of reflection spectra and number of contributing eigenmodes for both geometries, showing $2 \leqslant N_{\text{modes}} \leqslant 10$ active modes across the range of interest ($\lambda \approx [1.3\,\mu\text{m}, 2.0\,\mu\text{m}]$). The reflection spectra are marked with circles corresponding to the wavelengths considered in (a) and (b). (e) and (f) The first ten E_y mode profiles with the active modes highlighted in the corresponding color.

where $\Psi(z)$ is the complex-valued electric field at a given normalized depth $Z = z/z_0$, with $z_0 = 1\,\mu$m, W is a matrix containing the mode profiles, $\vec{\lambda}$ is a diagonal matrix containing each mode's complex-valued propagation constant, and $\vec{c}$ contains the amplitudes contained within each mode. In this representation, the real component of the ith propagation constant $\mathrm{Re}\{\lambda_i\}$ determines to what extent the ith mode decays. Because our metasurface layers are on the order of $1\,\mu$m, the number of contributing modes can be represented by $N_{\mathrm{modes}} = \arg(\mathrm{Re}\{\lambda_i\} < 1)$.

The values of $\mathrm{Re}\{\lambda_i\}$ at three wavelengths corresponding to local extrema in the reflection spectra are shown in Figs. 5(a) and 5(b) for the cylinder geometry and open-cylinder geometry. We then compute N_{modes} across all wavelengths for these two designs and plot them alongside the reflection spectra, shown in Figs. 5(c) and 5(d). In both solid- and open-cylinder metasurfaces we find $2 \leqslant N_{\mathrm{modes}} \leqslant 10$, depending on the wavelength of incident light. This indicates that both the number and amplitude within each active mode fluctuate over the region of broadband reflection. Moreover, we find the individual modal profiles are wavelength dependent [see Figs. 5(e) and 5(f)]. The total number of modes, the shape of the modes, and which modes are active at any given λ are thus a function of λ, making the task of orchestrating these modes to create the reflection spectra observed in Figs. 5(c) and 5(d) extremely onerous for all but the simplest geometries.

Previous research suggested it may be possible to orchestrate modal contributions to interfere with the incoming wave, resulting in nearly 100% reflection [56]. However, this was considered in a two-mode regime geometry, whereas the two systems we are considering convey the interplay of five or more at any given wavelength. Careful management of the interference of each mode within the metasurface to design a geometry with target performance is infeasible. Indeed, attempting such a solution as the design space increases to include not only the open-cylinder geometry but more complicated Bravais lattices, unit cells containing multiple meta-atoms, and different constituent materials is simply impossible. We conclude that the ANN-based solution to the inverse design of highly reflecting metasurface devices is both more achievable and more appropriate than an approach that requires deliberate handling of the active eigenmodes.

V. CONCLUSION

Metamaterials are a strong candidate for the design and fabrication of the next generation of multifunctional materials. Even with this limited simple design, the near-infinite number of available configurations makes exploring the parameter space a daunting task. However, by judiciously building and training ANNs, we showed a general pathway to solving the inverse design problem for metasurfaces. A PAD-trained PDPN was capable of advising open-cylinder metasurface designs by prescription. This network was then used to uncover an open-cylinder metasurface demonstrating high reflectivity $R \geqslant 99\%$ with $\Delta\lambda = 450$ nm of bandwidth, centered around a target wavelength $\lambda_0 = 1550$ nm. This performance exceeded the reflection bandwidths of comparable devices reported in previous studies. Modal analysis of this broadband design conveyed the complicated interaction of two to ten significant modes over the range of broadband reflection. From this, we claimed that ANNs provide a more efficient and general way for designing metasurfaces. Additionally, attempting to negotiate reflection properties by tracking modal interplay becomes infeasible for anything but the simplest cases. Considering other metasurfaces patterns is the immediate extension of this research, as well as considering different materials (other dielectrics, polymers), Bravais lattices, and complex unit cells.

While we saw sufficient predictive performance, the general pipeline for generating training data, the choice of neural network architecture, and the training parameters of the neural networks can all be subject to further examination. For example, different architectures could decrease the amount of training data necessary for convergence, or transfer learning methods could be implemented to encompass multiple types of metasurfaces into a singular ANN. All ANN-predicted designs can be further optimized by hybridizing existing optimization routines into the protocol or by directly factoring in manufacturing-related constraints. Finally, initial attempts utilizing modern electron-beam lithography and reactive ion etching techniques have shown the proposed open-cylinder geometries are realizable. Future work will involve scaling the manufacture process and assessing the optical performance of the fabricated metasurfaces.

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